

Fall 25 MATH 4100 Exam 2 Study Guide

The exam will cover lectures 9-22 to 10-30 and will be held during class on Monday 11-3. If you have testing accommodations, you will receive an email with further instructions.

Exam Structure: Questions will consist of multiple choice, true/false, computational and proof type questions. Partial credit will be given as appropriate. The exam will be closed note and you will be expected to memorize all relevant results with the exception of Graham-Schmidt.

List of Topics:

- Orthogonality
 - Graham-Schmidt
 - Orthogonal projection and Least squares
 - Linear functionals and inner product representations (Riesz Rep Theorem)
 - QR -factorization
- Determinants
 - Cramer's rule
 - Adjugate matrix
- Eigenvalues and eigenvectors
 - Characteristic and minimal polynomials
 - Similar matrices
 - Complex conjugate eigenvalues
- Diagonalization
 - Matrix exponential and solving $\mathbf{x}' = A\mathbf{x} - \mathbf{u}(t)$
 - Difference equations, $A\mathbf{x}_n = \mathbf{x}_{n+1}$, and linear recursive sequences

Suggested Exercises From Master Problem List: All types of computational problems are fair game. Here are a list of proof/conceptual problems to familiarize yourself with: 44, 50, 53, 57, 58, 63, 65, 66, 67, 69, 73cd